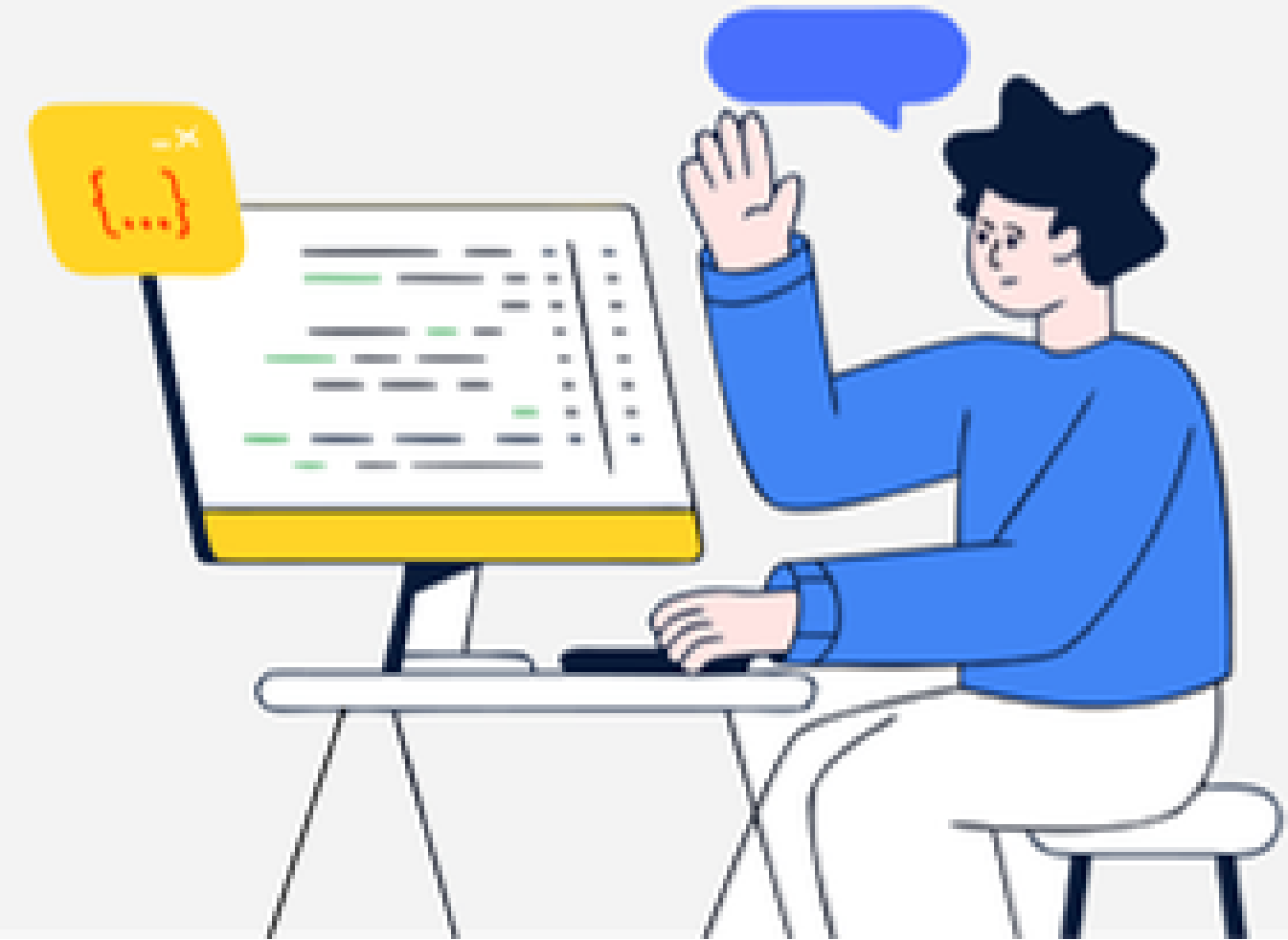


Solution Challenge



Team Details

- a. Team name: Sparkle Squad
- b. Team leader name: Pragga Mukherjee
- c. Problem Statement: Smart Resource Allocation

PROBLEM

What's broken?

- Data is scattered (paper, WhatsApp, files)
- Hard to analyze & act quickly
- Volunteers used inefficiently

Result:

- Delayed help
- Missed priorities
- Low impact

SOLUTION

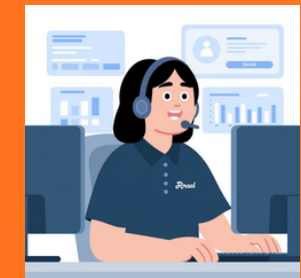
One Unified Platform that:



✓ Collects community data

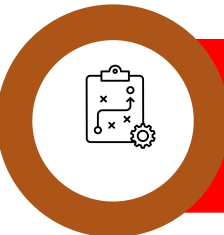


✓ Identifies urgent needs



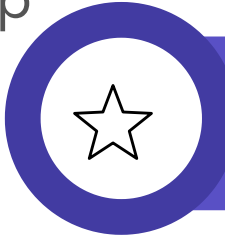
✓ Matches volunteers instantly

Aspect	Existing Solutions	Our Solution
System Approach	Data storage, manual use	Action-driven, real-time processing
Task & Decision Flow	Delayed, unstructured	Priority scoring + auto task generation
Volunteer Engagement	Poor coordination	Smart matching + micro-tasking



faster response time

Urgent needs are identified quickly, so help reaches people sooner.



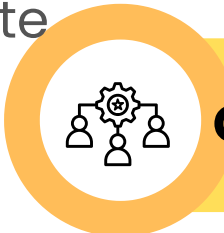
Higher participation

Micro-tasking allows more people to contribute, even with limited time.



Efficient volunteer use

Volunteers are assigned tasks based on skills and location, avoiding waste of effort.



Clear visibility of needs

Organizations always know where help is required most.

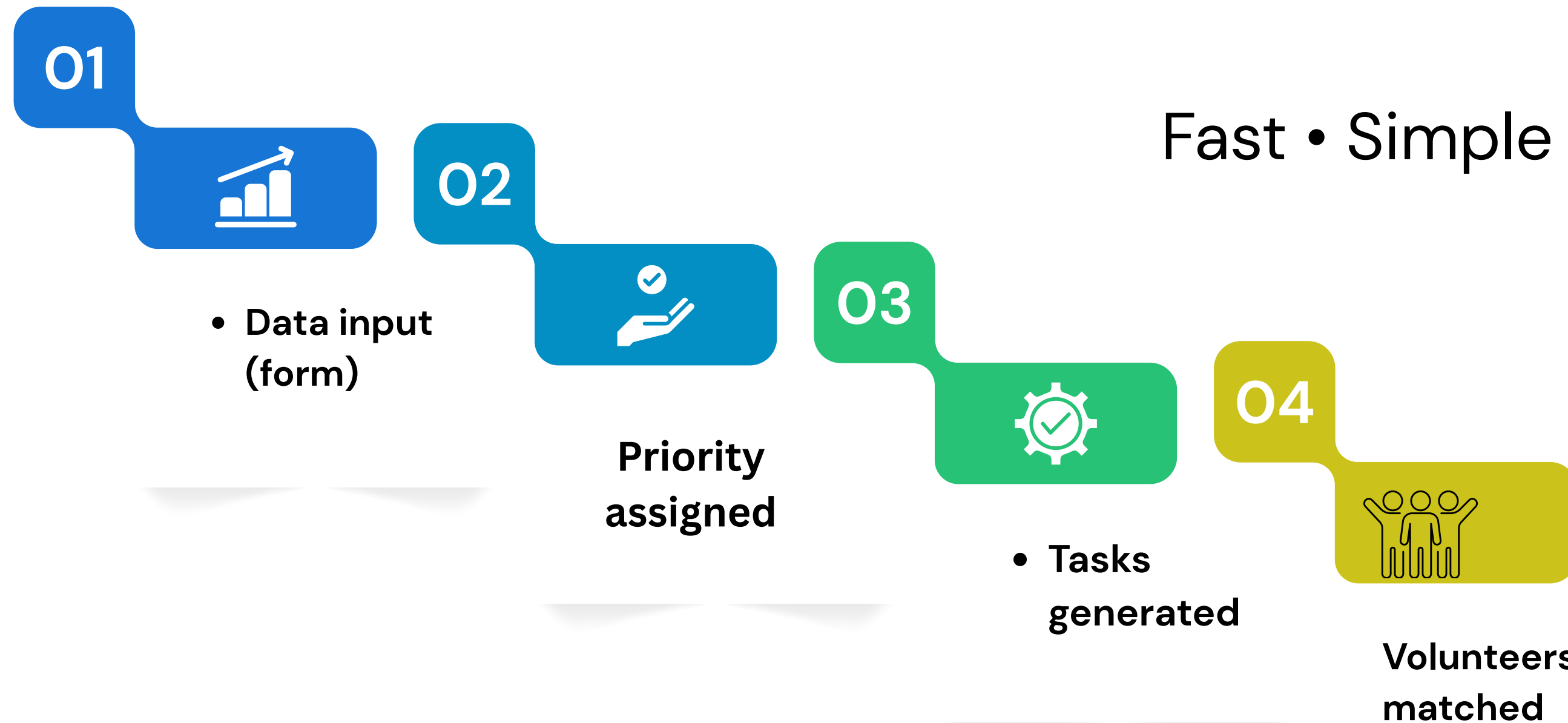
FEATURES

Big Problems → Small Tasks

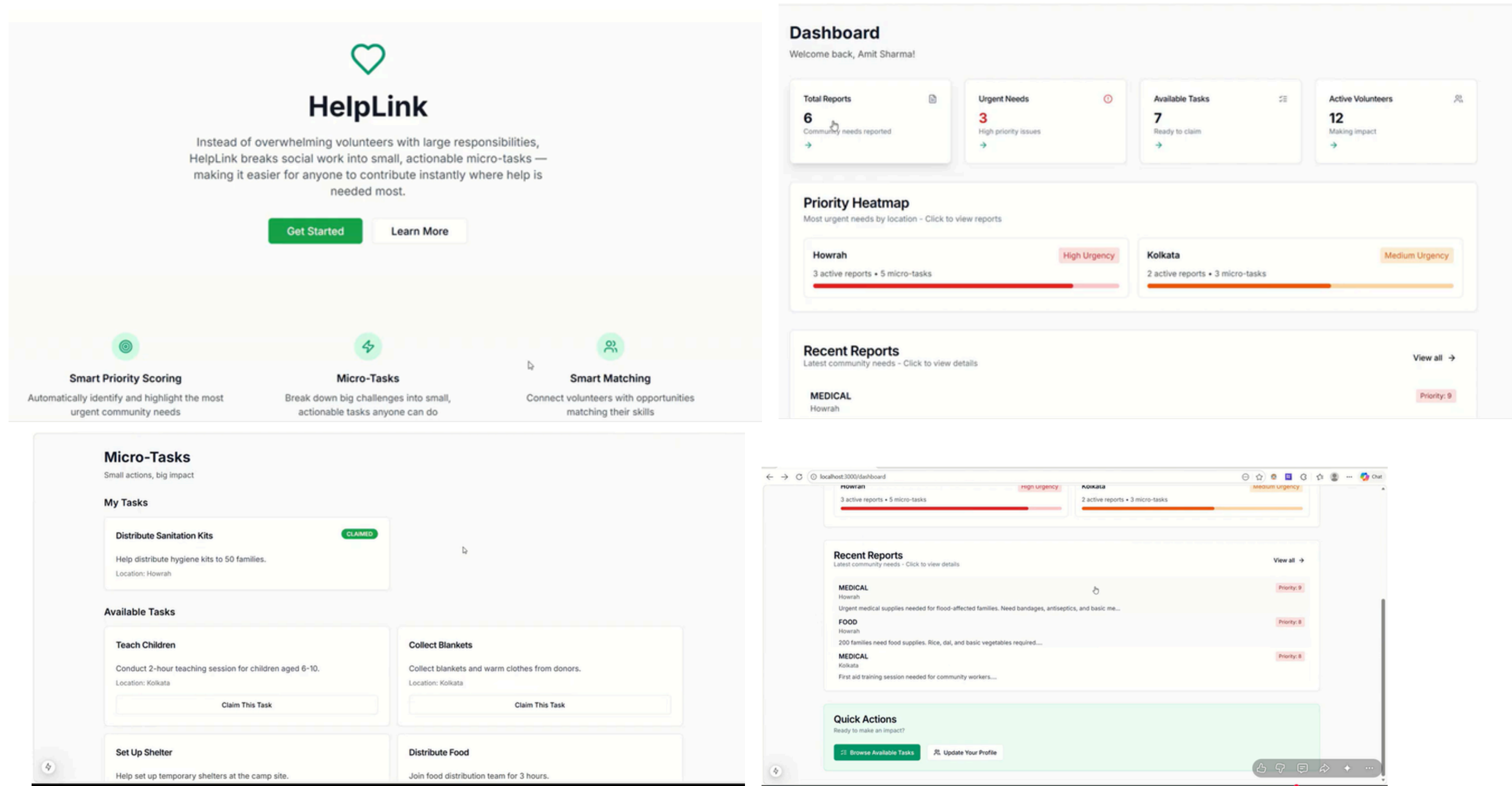


Unique
Feature:
Micro-Tasking

HOW IT WORKS

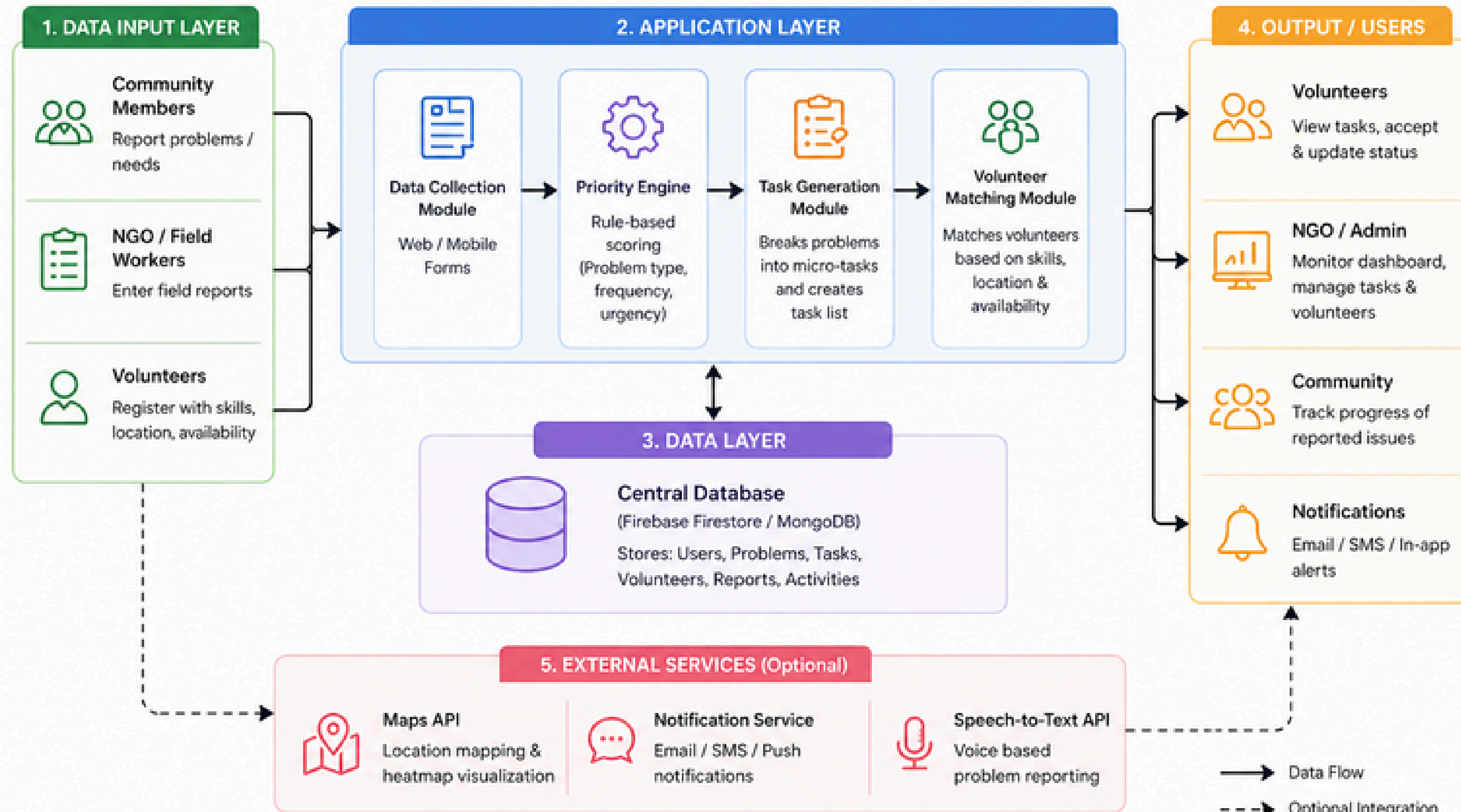


Wireframes/Mock diagrams of the proposed solution

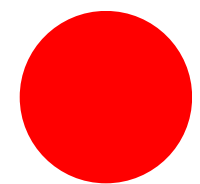


ARCHITECTURE DIAGRAM

Smart Resource Allocation System

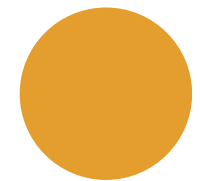


TECHNOLOGIES USED:



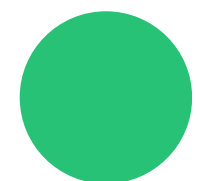
Frontend:

HTML, CSS, JavaScript
(or React)



Backend:

Node.js



Database:

MongoDB

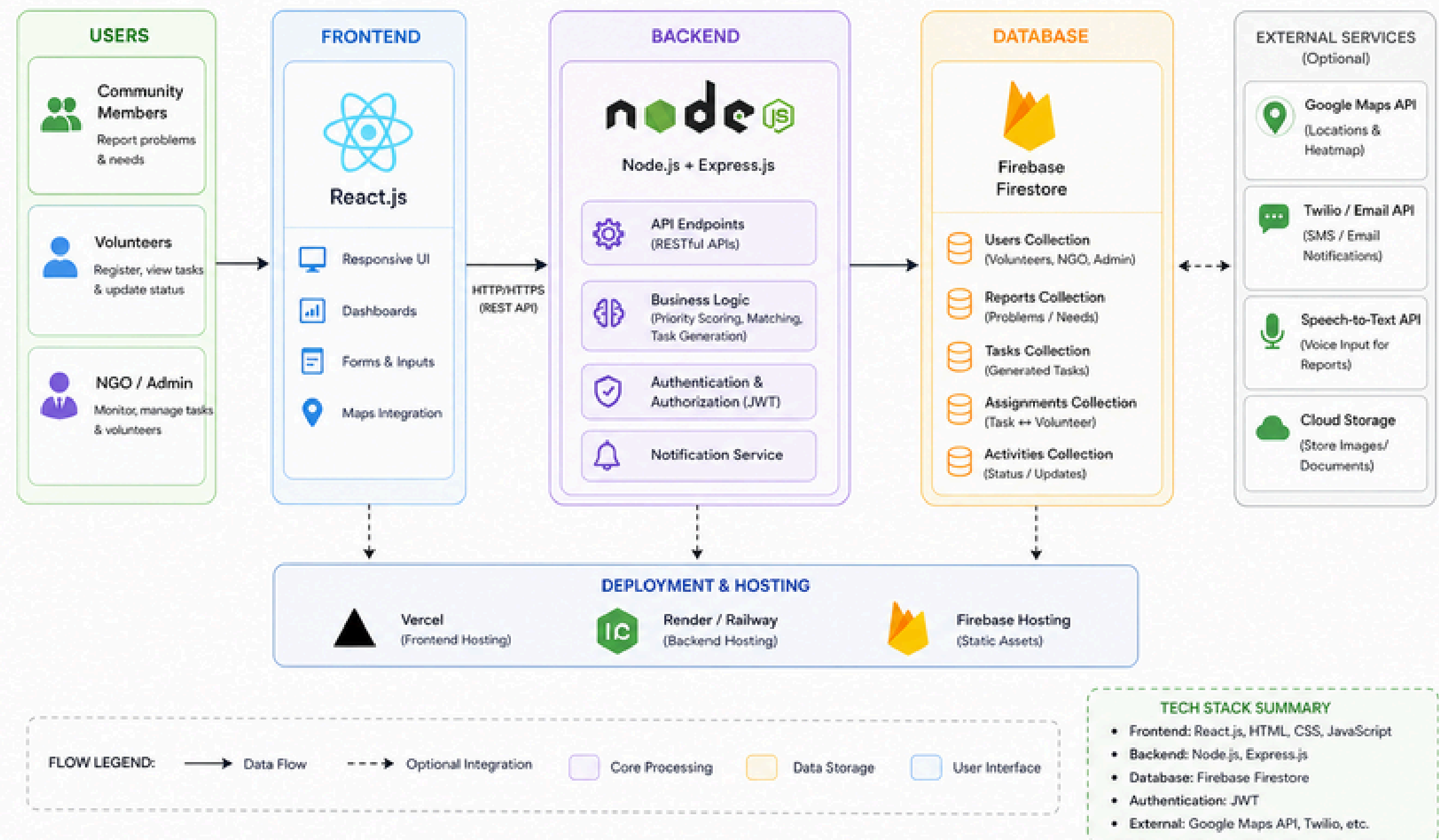


Priority Scoring (rule-based

Volunteer Matching (skill
+ location filtering)

Tech Stack Flow Diagram

Smart Resource Allocation System



Estimated implementation cost

Low-Cost & Scalable Solution

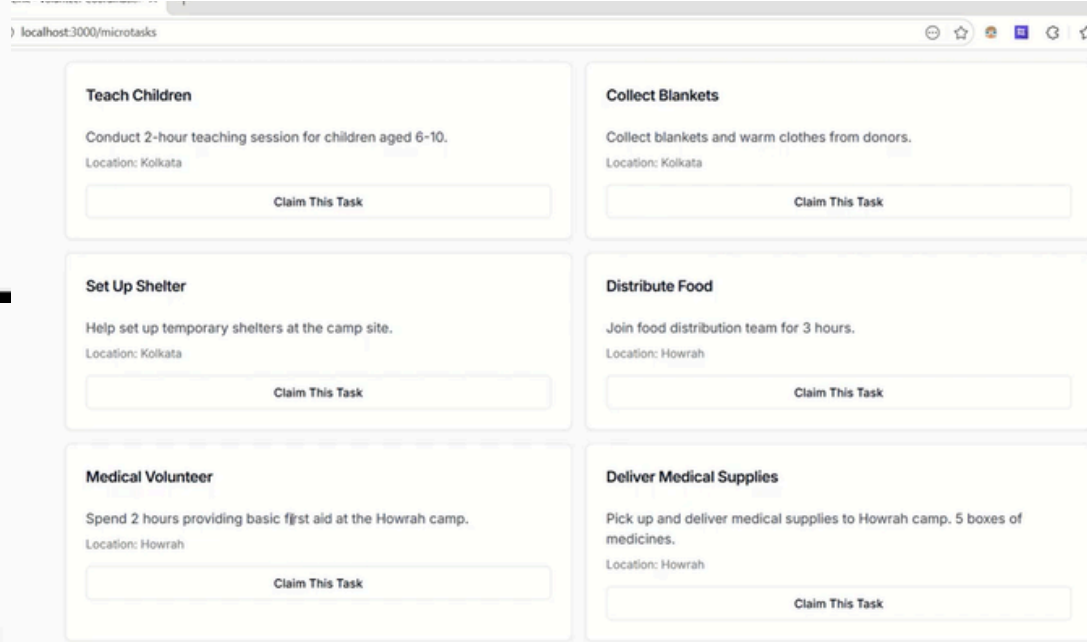
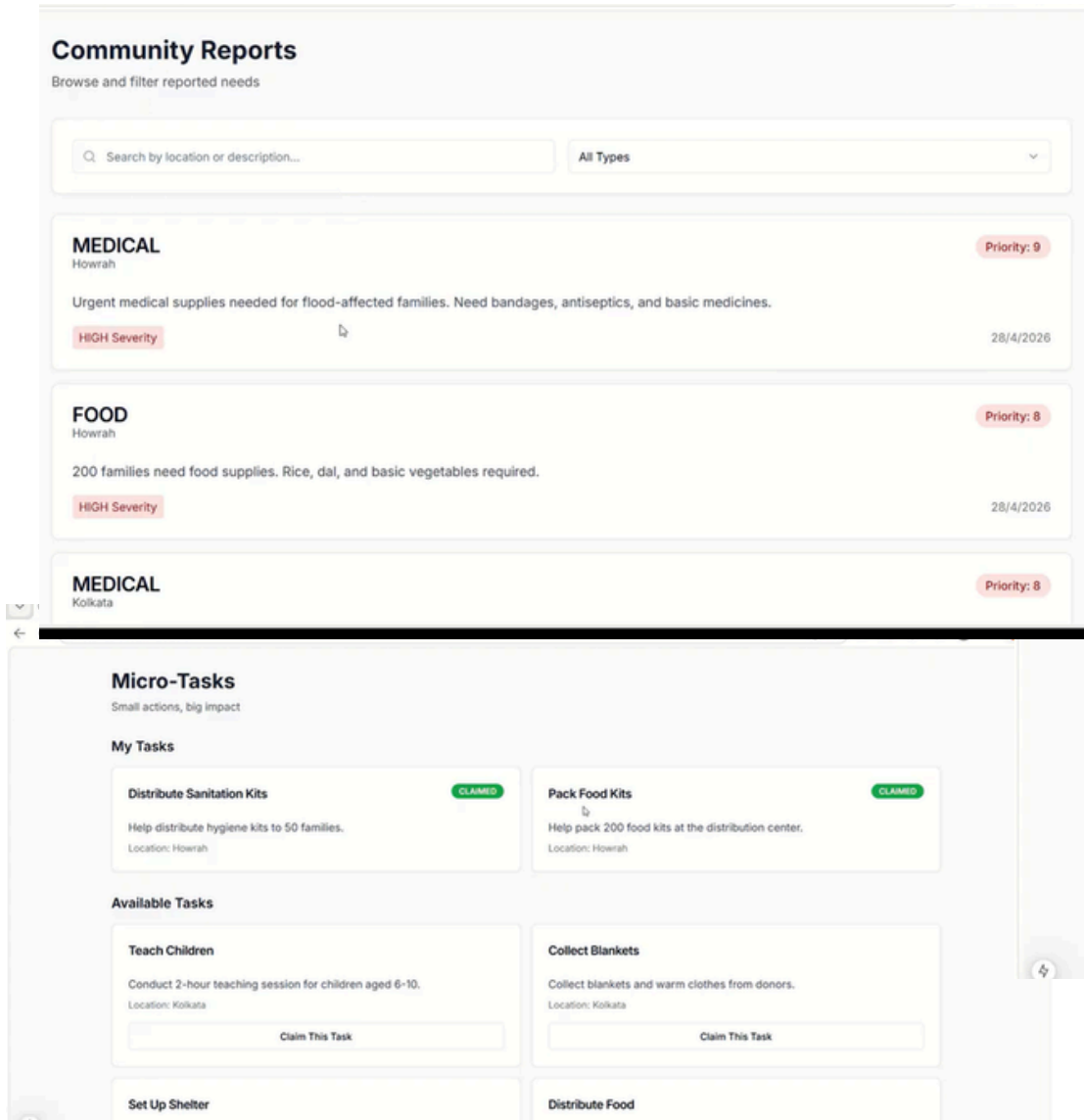
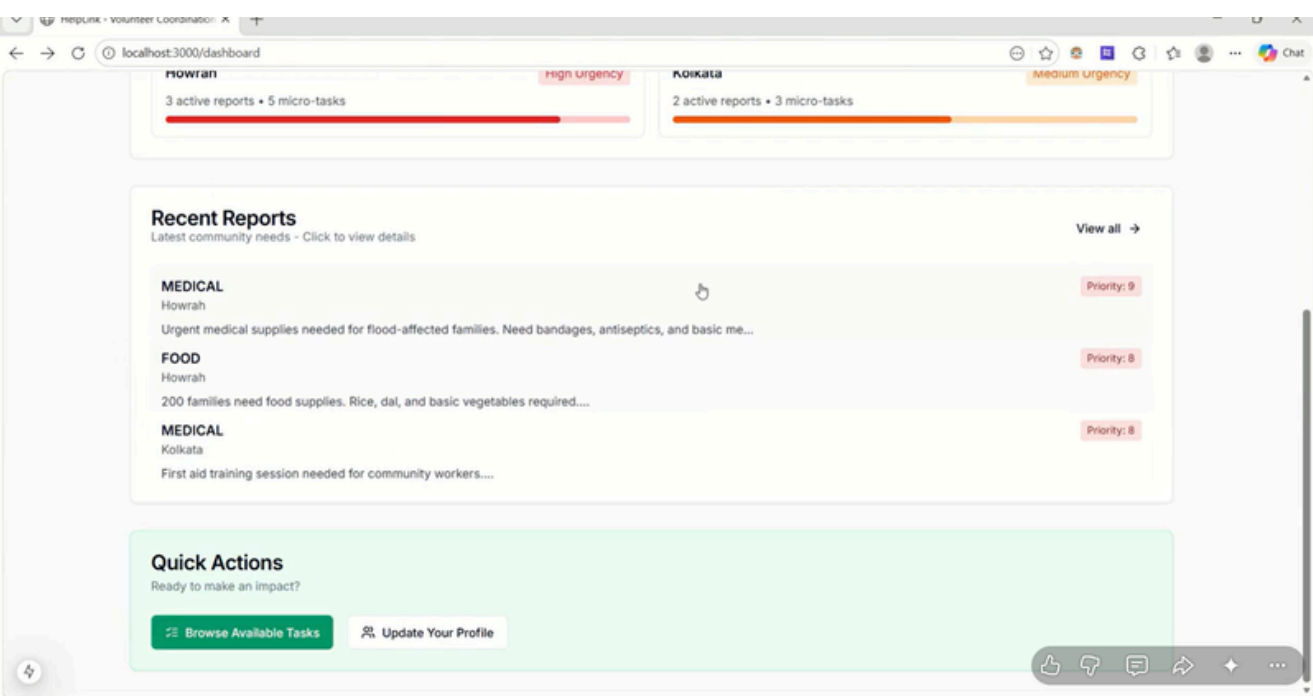
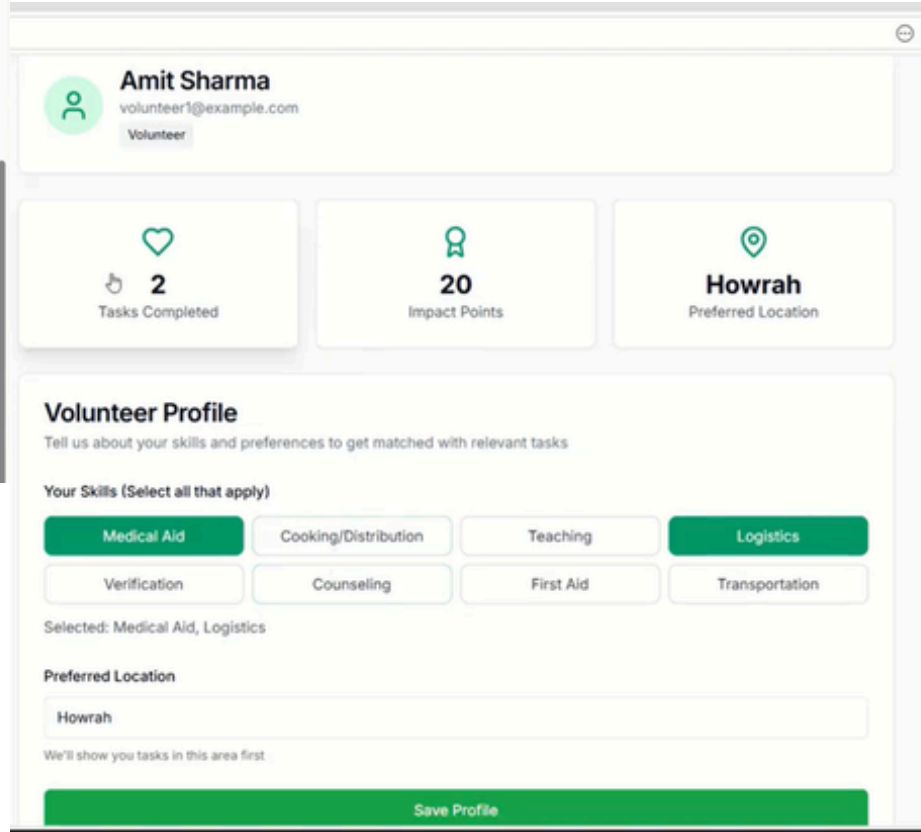
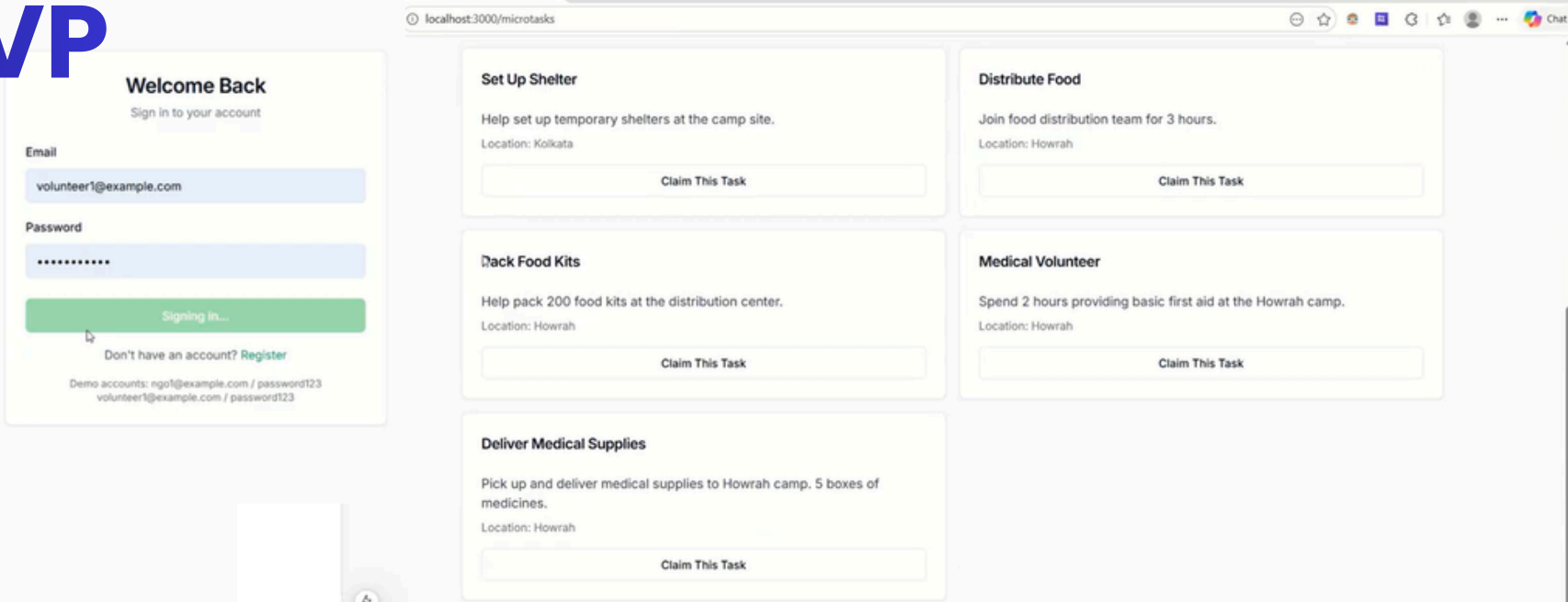
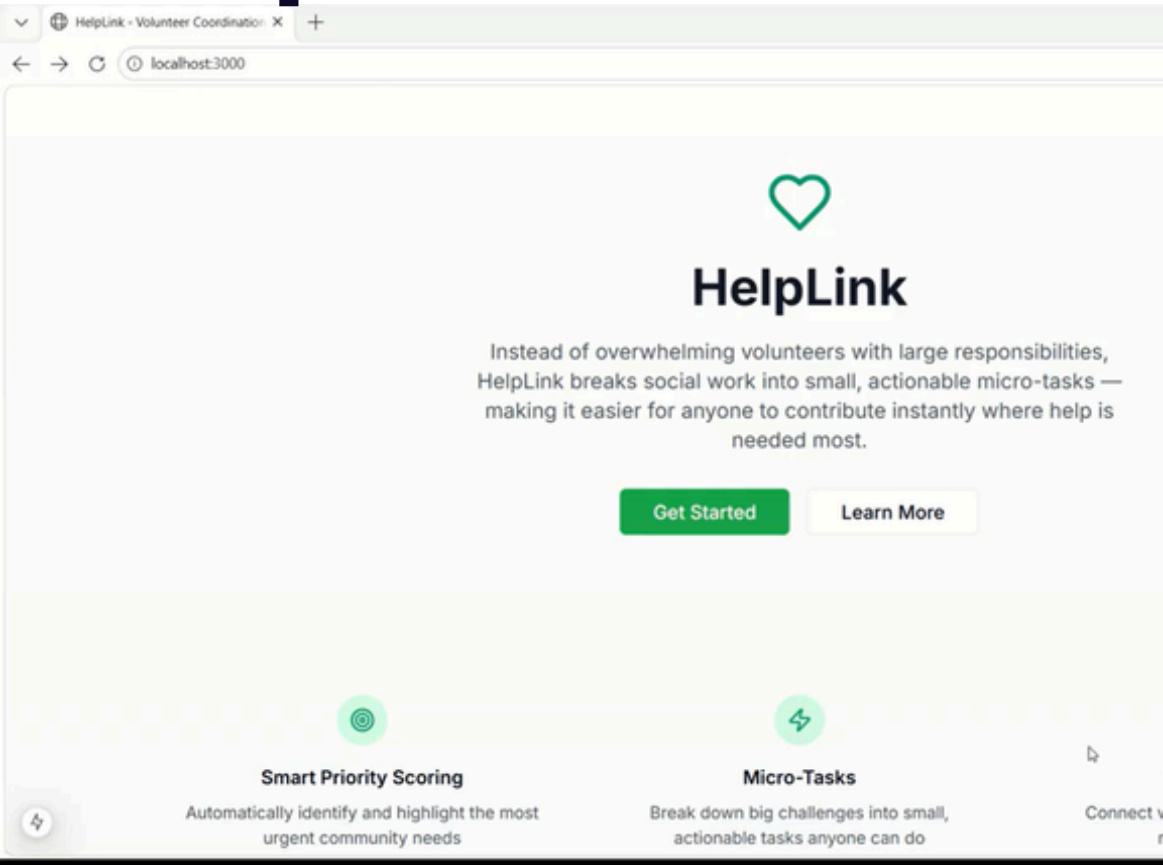
- **Development Cost:**
₹20,000 – ₹80,000
(basic version)
- Hosting & Storage: ₹500 – ₹2000/month
- Maintenance: Minimal
(small team required)

Why is the cost low?

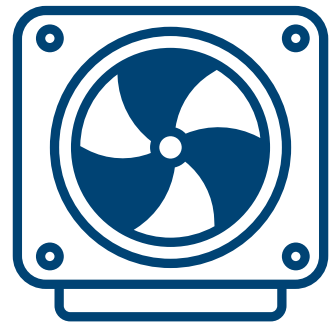
- Uses existing technologies & cloud platforms
- No need for expensive infrastructure
- Optimises existing resources instead of adding new ones

Cost decreases further using open-source tools

Snapshots of the MVP



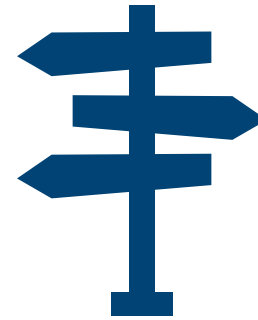
Planned Enhancements :-



- Mobile App Integration
- → Easy access for volunteers on-the-go



- AI-Based Priority System
- → Smarter and more accurate need identification



- Real-Time Location Tracking
- → Faster response and better coordination



- Instant Notifications
- → Alerts for urgent tasks and updates



- Multi-language Support
- → Wider accessibility across regions

Links:

1. GitHub Public Repository: <https://github.com/pragga9876/HelpLink.git>
2. Demo Video Link: <https://youtu.be/i7oM7gkR3F4?si=cayBRlajDIISxJO4>
3. MVP Link: <https://sparklesquadhelplink.vercel.app>
4. Working Prototype Link: <https://sparklesquadhelplink.vercel.app>



Solution Challenge



Thank You